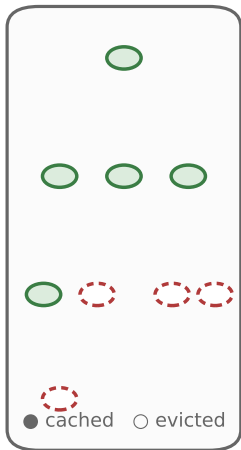


Recursive agent swarm



spawn

Bounded Infinity Cache (runtime)

Deterministic addressing
Cantor pairing $\kappa(\text{parent}, \text{child}) \mapsto \text{slot}$

Bounded cache — M fixed slots (full agent states)



Agent registry — $\mathcal{O}(N)$
(id, parent, depth, slot, status) per agent

Eviction manager (on full)

1. depth-priority select (evict deepest)
2. summarise φ (hierarchical)
3. fold into nearest cached ancestor

optional: Hilbert reorder for locality

query

Cache hit
 $\mathcal{O}(1)$ exact state

Cache miss
 $\mathcal{O}(\log N)$ ancestor-walk
reconstruction

registry $\rightarrow$ reconstruction